

A screen for synthetic genetic interactions with the *Saccharomyces cerevisiae* *hrq1ΔN* allele

Kennadi A. Shumaker ¹, Michael E. Kumcu,¹ Faith E. McDevitt ¹, Cody M. Rogers,^{1,2} Matthew L. Bochman ^{1,*}

¹Molecular and Cellular Biochemistry Department, Indiana University, Bloomington, IN 47405, United States

²Present address: Department of Biochemistry and Structural Biology, University of Texas Health Science Center at San Antonio, San Antonio, TX 78229, United States

*Corresponding author: Molecular and Cellular Biochemistry Department, Indiana University, Bloomington, IN 47405, United States. Email: bochman@iu.edu

The *Saccharomyces cerevisiae* *Hrq1* helicase is a functional homolog of the disease-linked human RECQL4 enzyme and has been used as a model to study RecQ4 helicase subfamily biology. Although the motor cores of *Hrq1* and RECQL4 are quite similar, these proteins display distinct N-terminal domains of unknown function. Do these domains facilitate species-specific activities by the two helicases, or do they serve common roles despite their differences in sequence and predicted structure? We probed these questions here by analyzing an N-terminal domains-truncated isoform of *Hrq1* (*Hrq1ΔN*) both in vitro and in vivo. We found that the *Hrq1* N-terminal domains houses a cryptic DNA binding site that is likely important for DNA repair because the *hrq1ΔN* allele phenocopies the DNA inter-strand crosslink sensitivity of *hrq1Δ*. Using synthetic genetic array analysis of *hrq1ΔN* crossed to the yeast *S. cerevisiae* single-gene deletion and temperature-sensitive allele collections, we also identified hundreds of synthetic genetic interactions, many of which are shared with previously characterized *hrq1* mutants. As with similar analyses of *hrq1Δ* and *hrq1*-K318A, our results suggest roles for *Hrq1* and its N-terminal domains in multiple physiological pathways that underpin genome integrity. Together, these data are guiding our ongoing efforts to understand the roles of *Hrq1* and RECQL4 in genome maintenance, which will help to explain why RECQL4 mutations cause disease.

Keywords: *Saccharomyces cerevisiae*; *HRQ1*; *hrq1ΔN*; DNA helicase; RecQ; RECQL4

Introduction

Mutations in the human RECQL4 DNA helicase cause three autosomal recessive diseases of genomic instability: Rothmund-Thomson (Ho et al. 2025), Baller-Gerold (Van Maldergem et al. 2007), and RAPADILINO syndromes (Vargas et al. 1992). Although all distinct, these diseases do include some overlapping pathologies, including bone morphology defects and predispositions to cancers (Balajee 2021; Xu et al. 2021). However, it is unclear how mutations in the helicase ultimately precipitate a disease state. Part of this quandary is the fact that RECQL4 is an evolutionary chimera, appearing to be a fusion of an essential DNA replication initiation factor and a RecQ family helicase (Liu 2010). The majority of the disease-linked alleles are located in the helicase portion of the protein (Larizza et al. 2010), presumably because perturbing the replication initiation function of RECQL4 would be embryonic lethal, but it is difficult to study the effects of these mutations without pleiotropically impacting DNA replication. Biochemical investigations of RECQL4 are similarly stymied by the protein's large size (~150 kDa) and natively disordered N-terminal domain (NTD), making recombinant protein purification difficult (Bochman et al. 2014; Keller et al. 2014; Rogers et al. 2017), as well as its strong annealing activity masking its DNA unwinding activity (Macris et al. 2006; Xu and Liu 2009; Keller et al. 2014).

To overcome these technical hurdles, we and others study the *Saccharomyces cerevisiae* *Hrq1* helicase, which is homologous to the RecQ portion of RECQL4 but lacks the confounding NTD related to

replication initiation (Barea et al. 2008; Bochman et al. 2014; Choi et al. 2014; Rogers and Bochman 2017; Rogers et al. 2017, 2020a, 2020b; Nickens et al. 2018, 2019, 2024; Sanders et al. 2020; Nickens and Bochman 2021; Luong et al. 2022; Nickens and Bochman 2022; Rajapaksha et al. 2022). To date, this work has drawn functional parallels between *Hrq1* and RECQL4, highlighting roles in genome maintenance processes such as DNA repair (Bochman et al. 2014; Rogers et al. 2017, 2020a, 2020c; Luong et al. 2022) and telomerase regulation (Bochman et al. 2014; Nickens et al. 2018, 2019, 2024; Nickens and Bochman 2021). However, like human RECQL4, *S. cerevisiae* *Hrq1* contains a large NTD of unknown function.

Are there cryptic nucleic acid binding sites in the *Hrq1* NTD as observed for RECQL4 (Keller et al. 2014)? Are there replication-independent roles shared by the *Hrq1* and RECQL4 NTDs? Are there species-specific functions? We sought to answer these questions here by characterizing an N-terminal truncation of residues 1 to 279 of *Hrq1* (*Hrq1ΔN*) both in vitro and in vivo. As part of the latter work, we performed synthetic genetic array (SGA) analysis of the *S. cerevisiae* *hrq1ΔN* allele by crossing it to the single-gene deletion collection (Giaever and Nislow 2014) and the temperature-sensitive (TS) collection (Kofoid et al. 2015) to generate a comprehensive set of double-mutant strains for analysis. This will allow us to compare synthetic genetic interactions with *hrq1ΔN* to those observed for *hrq1Δ* and *hrq1*-K318A (Rogers et al. 2020b; Sanders et al. 2020) to determine NTD-specific effects.

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Methods and materials

Media and other reagents

S. cerevisiae and *Escherichia coli* cells were grown in standard media at 30 and 37 °C, respectively, unless otherwise noted. Radioisotopes were purchased from Perkin–Elmer, and unlabeled ATP was from GE Healthcare. All restriction enzymes were from NEB, and all oligonucleotides were from IDT. All other reagents used were of the highest grade available. The *Saccharomyces* Genome Database was used extensively throughout this work for all things related to *S. cerevisiae* and its genes (Engel et al. 2025).

Recombinant protein production

Full-length *Hrq1* was over-expressed in baculovirus-infected insect cell culture and purified by Ni-affinity chromatography as previously described (Rogers et al. 2017). The *Hrq1* Δ N expression plasmid was created by PCR-amplifying the coding sequence for *Hrq1* amino acids 280 to 1,077 from pMB309 (Bochman et al. 2014) with oligonucleotides MB635 (CATGCATGTCATGAGCTGTCTCATCCACAATTTCGAAAAGGGTGCCAACTGGTCTCATCCACAATTTCGAAAAGGGTGCCAACTGGTCTCATCCACAATTTCGAAAAGGGTGGTGGAGCGGTGGTGGAGTTTGAAGTTTGTTCGAAGTCCAGGATCCCCGGAAGTATATCAGGGTATGGAACACG) and MB530 (CGTACGCTCGAGTATCTCTTTTAAATGATTTCATTTGTATGAGTATCGTCTTTCGTAGC), digesting the PCR product with BspHI and XhoI, and ligating it into the NcoI and XhoI sites of pET21d (Novagen) to create pMB296. The *Hrq1* NTD expression plasmid was generated by PCR-amplifying the coding sequence for *Hrq1* amino acids 1 to 279 from pMB309 using oligonucleotides MB653 and MB1032 (CGATCGATCTCGAGTGAAGTCAAAA CAGAGACCTTTATACTTCGC), digesting the PCR product with BspHI and XhoI, and ligating it into the NcoI and XhoI sites of pMB131 (Paeschke et al. 2013; Andis et al. 2018) to create pMB426. These plasmids were transformed into Rosetta 2(DE3) pLysS cells (Novagen), and the *Hrq1* truncations were over-expressed using the high-density autoinduction method (Studier 2005). Both proteins were purified using Strep-Tactin Sepharose and Ni-affinity chromatography as previously described for full-length *Hrq1* (Bochman et al. 2014). Additional details concerning the cloning of the expression constructs and protein purification schemes are available upon request.

Transmission electron microscopy

To prepare negatively stained grids, 4 μ L of protein solution (~20 ng/ μ L) was applied to glow-discharged 300-mesh carbon-coated copper grids (Electron Microscopy Sciences). Excess solution was removed by blotting with filter paper, and subsequently, the grid was stained with 0.75% uranyl formate (Electron Microscopy Sciences) and allowed to air dry. The grids were then imaged under low-dose conditions (≤ 20 e $^-/\text{\AA}^2$) using a JEOL 300-kV 3200FS transmission electron microscope with the energy filter operated at an energy slit of 20 eV to minimize potential beam-induced damage to the protein samples. The digital images were collected using a Gatan 4k $\times$ 4k CCD camera at a nominal magnification of 80,000 $\times$, yielding a 1.5 $\text{\AA}$ /pixel resolution in the final images.

In silico protein analyses

AlphaFold 3 models were rendered on the Google DeepMind AlphaFold Server (alphafoldserver.com) using default parameters, unless otherwise noted. The protein sequence for *S. cerevisiae* *Hrq1* from the S288c genetic background was obtained from the *Saccharomyces* Genome Database (yeastgenome.org). The *Schizosaccharomyces pombe* *Hrq1* sequence was from PomBase

(https://www.pombase.org/gene/SPAC23A1.19c). The other RecQ4 family helicase sequences were retrieved from UniProt (https://www.uniprot.org/). See Supplementary Table 1 for additional details. Structures were visualized and analyzed in Chimera X (https://www.cgl.ucsf.edu/chimerax). The disorder analyses utilized online PrDOS (https://prdos.hgc.jp/cgi-bin/top.cgi), DEPICTER (https://biomine.cs.vcu.edu/servers/DEPICTER2/), DISOPRED3 (https://github.com/psipred/disopred), and IUPred (https://iupred2a.elte.hu/) servers with default parameters.

Diepoxybutane sensitivity assay

Wild-type MBY49 (*MATa ura3-52 lys2-801_amber ade2-101_ochre trp1 Δ 63 his3 Δ 200 leu2 Δ 1 hxt13::URA3*) (Paeschke et al. 2013; Bochman et al. 2014) and its *hrq1 Δ* (MBY161, *hrq1::his3MX6*), *hrq1-K318A* (MBY346, *hrq1::hrq1-K318A(natMX)*), and *hrq1 Δ N* (MBY462, *hrq1::hrq1 Δ N(natMX)*) derivatives were tested for diepoxybutane (DEB) sensitivity essentially as described (Bochman et al. 2014). Briefly, the cells were grown overnight in 1% yeast extract, 2% peptone, and 2% dextrose (YPD) medium at 30 °C with aeration, diluted to an optical density at 600 nm of 1 with sterile water, and then 10-fold serially diluted likewise to 10 $^{-4}$. Then, 5 μ L of each dilution was spotted onto plates either containing or lacking DEB (Sigma) at the indicated concentration. The plates were incubated in the dark at 30 °C for ~2 d before being imaged with a flat-bed scanner. DEB sensitivity assays were performed in triplicate with representative scans presented in Fig. 3. All strains were derived from the wild-type YPH genetic background (Sikorski and Hieter 1989) by standard methods. Details of the strain constructions are available upon request.

NADH-coupled ATPase assay

ATPase reactions were performed in ATPase buffer (25 mM Na-HEPES [pH 8.0], 5% glycerol, 50 mM NaOAc [pH 7.5], 150 μ M NaCl, and 0.01% NP-40 substitute) including the following reagents: 5 mM ATP (pH 7.0) (DOT Scientific), 5 mM MgCl $_2$, 0.5 mM phosphor(enol)pyruvic acid (Sigma), 0.4 mM NADH (MP Biomedicals, LLC), 5 U/mL rabbit pyruvate kinase (Roche), and 8 U/mL lactate dehydrogenase from oyster (Sigma). Unless otherwise stated, the helicase concentration was 10 nM in all reactions, and poly(dT) $_{50}$ single-stranded (ss)DNA was used at 1 μ M. To minimize possible confounding effects caused by secondary structures that the oligonucleotides may form, the substrates were diluted to their working concentrations, boiled, and snap-cooled prior to addition to the ATPase assays. Absorbance at 340 nm was read at 37 °C in 96-well plates using a BioTek Synergy H1 microplate reader. Absorbance readings were converted to ATP turnover based on NADH concentration. It was assumed that 1 μ M NADH oxidized is proportional to 1 μ M ATP hydrolyzed.

DNA binding

DNA binding was measured using electrophoretic mobility shift assays (EMSAs). The proteins were incubated at the indicated concentrations with 0.1 nM radiolabeled DNA for 30 min at 30 °C in binding buffer (25 mM Na-HEPES [pH 8.0], 5% glycerol, 50 mM NaOAc [pH 7.6], 150 μ M NaCl, 7.5 mM MgOAc, and 0.01% Tween-20). Protein–DNA complexes were separated from unbound DNA on 8% 19:1 acrylamide:bis-acrylamide gels in 1 $\times$ TBE (90 mM Tris–HCl [pH 8.0], 90 mM boric acid, and 2 mM EDTA [pH 8.0]) buffer at 10 V/cm. Gels were dried under vacuum and imaged using a Typhoon 9210 Variable Mode Imager. DNA binding was quantified using ImageQuant 5.2 software. The DNA substrate was made by 5'-end labelling a poly(dT) 30mer oligonucleotide with T4 polynucleotide kinase (T4 PNK; NEB) and γ [32 P]-ATP.

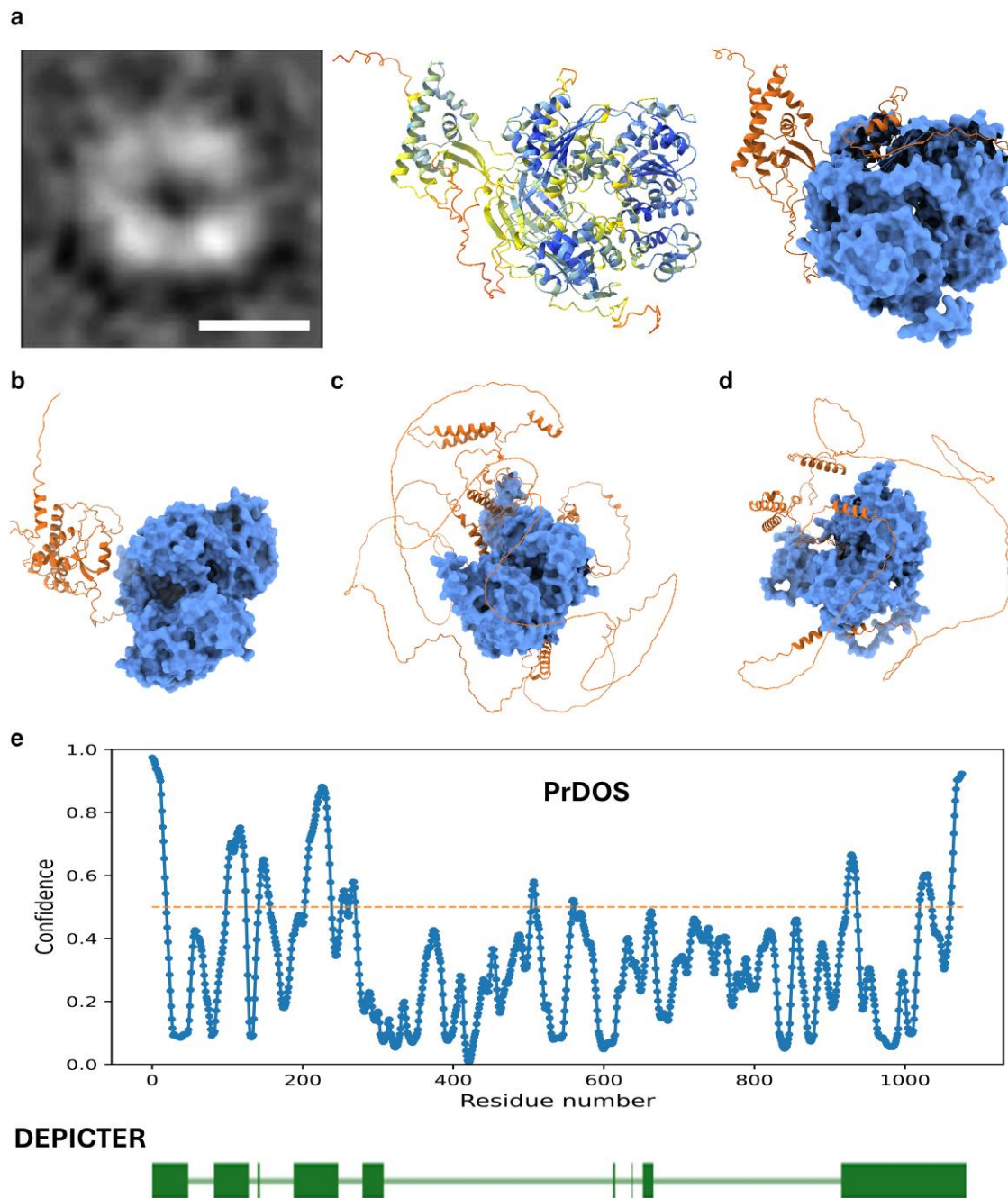


Fig. 1. Structural considerations for the NTD of *Hrq1*. a) TEM of a recombinant *Hrq1* particle (left; scale bar = 100 Å), an AlphaFold 3 prediction of the *Hrq1* structure colored by confidence score (middle; cooler colors = more confident), and the same predicted structure depicted with the helicase core space filled and the NTP (aa 1 to 279) in ribbon format (right). These images depict a toroidal structure with a central void, but density corresponding to the NTD is missing in the micrograph, suggesting that the NTD is highly conformationally flexible or intrinsically disordered. AlphaFold predictions for *S. pombe* *Hrq1* (b), *D. melanogaster* RecQ4 (c), and human RECQL4 (d) are also shown. e) Disorder prediction for *Hrq1* indicates a high confidence for native disorder in the NTD. The PrDOS server (top) predicts several disordered regions in the NTD (residues with values above the dotted line). The DEPICTER server (bottom) also predicts native disorder in the NTD (green boxes), as well as a largely disordered C-terminus.

Labelled oligonucleotides were separated from free label using illustra ProbeQuant G-50 micro columns (GE Healthcare) following the manufacturer's instructions.

SGA analysis

To generate the *hrq1ΔN* SGA query strain, the natMX6-tagged allele was PCR-amplified from MBY462 using oligonucleotides MB527 (GTGAATTGCTCAGAAGAGAAAGGCATACCGTC) and MB528 (CTGTGCATCAACAAGGTGACAGAATGTTGATG). This

PCR product was then used to transform strain Y8205 (*MATa can1Δ ::STE2pr-Sp_his5 lyp1Δ ::STE3pr-LEU2 his3Δ1 leu2Δ0 ura3Δ0*) (Tong et al. 2001, 2004). The replacement of the endogenous *HRQ1* gene with the *hrq1ΔN* allele was verified by PCR analysis using genomic DNA and oligonucleotides MB527 and MB528 that anneal to regions up- and downstream of the *HRQ1* locus. The confirmed *hrq1ΔN* strain was named MBY641.

SGA analysis of the *hrq1ΔN* mutant was performed at the University of Toronto using previously described methods (Tong

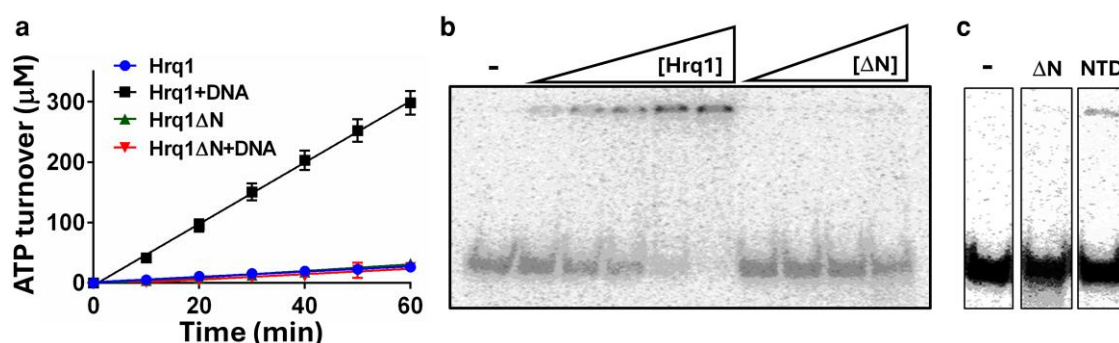


Fig. 2. The *Hrq1* NTD affects ATP hydrolysis and DNA binding. a) *Hrq1* and *Hrq1*ΔN (10 nM) have low basal levels of ATPase activity. The addition of 1 μM poly(dT) ssDNA stimulates *Hrq1* ATP hydrolysis (*Hrq1* + DNA) but has no effect on *Hrq1*ΔN (*Hrq1*ΔN + DNA). The plotted points are the averages of ≥ 3 independent experiments, and the error bars are the standard deviation. b) Gel shift analysis of *Hrq1* (0.2 to 5 nM) and *Hrq1*ΔN (0.5 to 5 nM) binding to poly(dT) ssDNA. *Hrq1*ΔN is largely devoid of ssDNA binding activity, consistent with ssDNA failing to stimulate its ATPase activity. c) Example gel shift data showing that the *Hrq1* NTD in isolation displays a weak ssDNA binding activity. –, no protein added; ΔN, 10 nM *Hrq1*ΔN added; and NTD, 10 nM *Hrq1* NTD added.

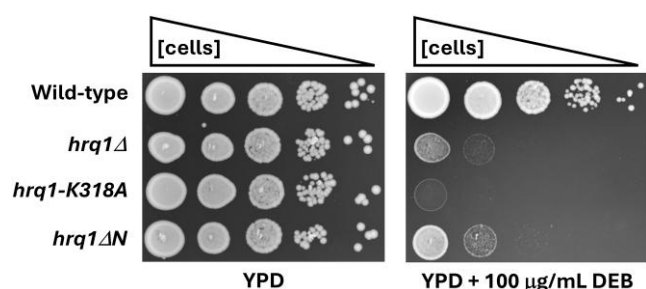


Fig. 3. Cells expressing *Hrq1*ΔN phenocopy the DEB sensitivity of *hrq1*Δ cells rather than the stronger sensitivity of *hrq1*-K318A cells. The plate images show the growth of 10-fold serial dilutions of the indicated strains at a uniform density incubated for 48 h at 30 °C.

et al. 2001, 2004). MBY641 was crossed to both the *S. cerevisiae* single-gene deletion collection (Giaever and Nislow 2014) and the TS alleles collection (Kofoed et al. 2015) to generate double mutants for analysis. For these screens, the control strain (Y8835) contained the NatMX marker inserted into the benign *ura3* locus (*MATa ura3Δ::natMX4 can1Δ::STE2pr-Sp_his5 lyp1Δ his3Δ1 leu2Δ0 ura3Δ0 met15Δ0 LYS2+*). Quantitative scoring of the genetic interactions was based on colony size. The SGA score measures the extent to which a double-mutant colony size deviates from the colony size expected from combining two mutations together. The data include both negative (putative synthetic sick/lethal) and positive interactions (potential epistatic or suppression interactions) involving *hrq1*ΔN. The magnitude of the SGA score is indicative of the strength of the interaction. Based on statistical analysis, it was determined that a default cutoff for a significant genetic interaction is $P < 0.05$ and SGA score $> |0.08|$. It should be noted that only top-scoring interactions were confirmed by remaking and re-analyzing the double mutants by hand. The full results of the SGA analyses are found in Supplementary Table 2. Lists of shared and unique interactions between *hrq1*Δ, *hrq1*-K318A, and *hrq1*ΔN are found in Supplementary Tables 3 and 4.

Confirmation of top SGA hits

The top five positive and negative interactors with *hrq1*ΔN from the single-gene deletion and TS arrays were reanalyzed by hand to confirm their phenotypes. Briefly, the SGA query strain MBY641 (Nat^R) was mated to *MATa* tester strains from the arrays (Kan^R), sporulated, and then analyzed by random spore analysis

(Lichten 2014), spot dilution (Andis et al. 2018), and/or growth curve (Ononye et al. 2020; Sausen and Bochman 2021) assays of Nat^R Kan^R spore clones compared to the parental single-mutant strains and wild-type. Examples of these validations are shown in Supplementary Fig. 3.

Statistical analysis

Data were analyzed and graphed using GraphPad Prism 6 software. The reported values are averages of ≥ 3 independent experiments, and the error bars are the standard deviation. P-values were calculated as described in the figure legends, and we defined statistical significance as $P < 0.01$.

Results and discussion

The Hrq1 NTD is predicted to contain native disorder

The N-terminus of human RECQL4 is intrinsically disordered yet contains important functional elements (Keller et al. 2014). Although the *Hrq1* NTD does not display similarity to the RECQL4 NTD at the primary sequence level, we hypothesized that the native disorder might be conserved rather than the amino acid sequence. We base this conjecture on the fact that negatively stained TEM images of single *Hrq1* particles have a multilobed “C-” or “O-shaped” structure (Fig. 1a; Rogers et al. 2017) that is roughly consistent with the size and shape of the AlphaFold-predicted structure of *Hrq1* minus its N-terminal ~279 amino acids (Fig. 1a). In addition to this portion of the structure being predicted to include long stretches of random coil, the confidence of the prediction is lower in the NTD compared to the helicase core of the enzyme, consistent with a propensity for molecular motion or disorder (Abramson et al. 2024). This predicted combination of intrinsic disorder and low-confidence secondary structural elements is also displayed by the AlphaFold model of the *Schizosaccharomyces pombe* *Hrq1* (Fig. 1b), though in the *Drosophila melanogaster* RecQ4 (Fig. 1c) and human RECQL4 (Fig. 1d) homolog models, the amount of random coil is increased. Supplementary Fig. 1 shows additional AlphaFold models of RecQ4 family helicases and their NTDs.

To further probe the predicted structure of the *Hrq1* NTD, we submitted the *Hrq1* sequence to the PrDOS (Ishida and Kinoshita 2007) and DEPICTER (Barik et al. 2020) protein disorder prediction servers for analysis. As shown in Fig. 1e, both algorithms predict several regions of native disorder in the *Hrq1*

Table 1. Synthetic genetic interactions with *hrq1ΔN*.

Negative interactions ^a	Positive interactions
Gene deletions YSC83, SLX9, GIC1, MRM2, SPO16, UTH1, TOM70, SWC5, PHO5, YME1, YJU3, MDM35, RPL6B, YUR1, STE50, MUP3, HAP2, HAP3, PCK1, BEM1, DAK1, TIM18, ASK10, UBP1, MBR1, TCO89, RAD4, KCH1, SWI4, AIM4, YJL213W, ENT3, MIR1, PET127, CDC73, GPR1, ELM1, RAD14, IRA2, BDF2, PHB2, MLS1, PMA2, VAM6, CSE2, MNN11, BTS1, IMG2, YPR084W, YML090W, TGS1, MDH3, GRR1, HTD2, SPO20, COX10, YJL175W, IXR1, PEF1, YHK8, QCR9, KGD2, YOS9, KAP114, PET494, UPS1, YMR245W, OAR1, LTE1, RPL43B, PAC10, YBR238C, CTI6, LPD1, FPR2, LIP5, LDB19, YJR120W, SAP190, SIN3, COX23	INP2, RPS27B, OLA1, DGA1, YML012C-A, YLR338W, AAH1, SWI6, BNA6, DIA2, ATG22, TSR2, ERG3, UBS1, VRP1, ILM1, ADE2, ACB1, PEX12, TSR3, VPS9, ARF1, RPO41, NHP10, YHM2, SIM1, MSH4, SHE4, OST4, VPS4, TRM44, SPR6, PDE1, RPN4, ART5, VID30, YGL024W, EDC1, CSM1, ALT1, YBL094C, SAC1, YHR180W, NDE1, ARC1, YAR1, YDL062W, FEN1, YBR016W, YMR031W-A, GRX2, MSB4, YMR294W-A, IES5, YJL206C-A, EOS1, VAM7, PKP2, HXT2, KAR5, CAT2, MFT1, MKS1, RTG1, ICE2, ARA2, RPS9B, RTG3, RTG2, YGL214W
Temperature-sensitive alleles yef3-f650s, mob2-8, efb1-4, mob2-38, taf2-1, mob2-36, mob2-14, mob2-28, cse2, mob2-34, med7-163, cdc31-5, mob2-40, mob2-20, mob2-11, mps1-1, mob2-19, smc2-8, mob2-22, pol1-17, yhr172w-ph, ycs4-1, smd1-1, rlp7-ph, yhc1-2, tub2-443, dbp5-1, rox3-182, pse1-34, sec63-1, mcm1-ph, cdc21-1, mas2-10, arp3-g15c, mob2-24, eaf7, puf6, spt3, htd2, apq12, bbb1-1, cdc20-1, nup57-e17, lst8-6, yil061c-ph, neo1-2, cof1-5, pob3-q308k, ndc1-4, ykr022c-ph, lip5, ipi1-ph, rrp4-1, yil106w-ph, sgf73, dfr1-td, fip1-ph, srb6-ph, lst8-15, tfq2-ph, slm3, nab3-11, pam16-3, pre7-ph, lge1, cct4-1, swd2-ts1, spc29-3, stb5, nut2-ts, dop1-1, arp3-f306 g, pth1, cmd1-3, rps22b, qcr2	gpi12-ph, myo2-14, sts1-ph, ymr134w-ph, slx9, sec24-2, sec27-1, sec23-1, gtr1, pds5-1, ies5, act1-129, vam6, ykt6-ph, srv2-2, orc2-4, sed5-1, act1-112, rpf2-ph, stt3-2, stu2-10, pfy1-4, rvb2-ph, bet3-ph, orc1-ph, sur1, act1-2, yhc1-6, ies1, stt3-1, gpi10-ph, gaa1-ts, rpc40-u78r, pbn1-ph, smc4-1, nhp10, sda1-2, brn1-9, spt5-194, rps5-sm1, act1-155, nse4-ph, act1-3, ebp2-1, cdc1-4, dpb11-1, cdc11-5, arp3-d11a, rrm7-ph, sfi1-7, fal1-1, spt6-14, cdc2-7, cdc48-1, ydl003w-ph, apc11-22, arp3-g302y, las17-1, mps3-1

^a Negative genetic interactors are listed from largest absolute value of the SGA score to the smallest, but positive interactors are listed from the smallest to the largest.

NTD (as well as the C-terminus), though the predictions do not match in the number or length of the disordered motifs. A similar analysis using the DISOPRED (Jones and Cozzetto 2015) and IUPred (Erdos and Dosztanyi 2024) servers was also performed but with similarly inconsistent results (Supplementary Fig. 2). The DISOPRED prediction closely resembles that of PrDOS (Fig. 1b and Supplementary Fig. 2a), but the IUPred algorithm predicts intrinsic disorder only at the very N-terminal portion of the Hrq1 NTD (Supplementary Fig. 2b).

Based on the structure prediction (Fig. 1a) and three of four disorder predictors (Fig. 1e and Supplementary Fig. 2a), it appears likely that the N-terminus of Hrq1 contains regions of native disorder but also folded secondary structures. The NTD of the second *S. cerevisiae* RecQ family helicase Sgs1 is somewhat similar, with long stretches of disordered amino acids interrupted by short motifs that dynamically sample folded and unfolded space, as indicated by NMR analysis (Kennedy et al. 2013). Importantly, these portions of Sgs1 are necessary for its *in vivo* function. Although we only consider mutation of Hrq1 in the context of a full NTD truncation herein, future research should be devoted to disrupting the predicted secondary structures in the Hrq1 NTD to determine their physiological roles as a finer-grained approach.

The NTD affects Hrq1's ATPase and DNA binding activities

To investigate the biochemical roles of the Hrq1 NTD, we generated recombinant full-length wild-type and Hrq1ΔN proteins. We previously demonstrated that Hrq1 and RECQL4 are DNA-stimulated ATPases, whose activities are impacted by DNA sequence and structure (Rogers and Bochman 2017; Rogers et al. 2017). Here, we tested the Hrq1ΔN preparation for ATPase activity and found that it was identical to that of full-length Hrq1 (Fig. 2a), suggesting that the helicase core of Hrq1ΔN folds properly in the absence of the NTD. However, the ~5-fold stimulation of wild-type Hrq1 ATPase activity by ssDNA is lacking in the Hrq1ΔN preparation. Addition of the purified NTD *in trans* to the Hrq1ΔN ATPase assays had no effect, nor did the NTD display ATP hydrolysis on its own (data not shown). Taken together, these data

suggest that the Hrq1 N-terminus has an important role in coordinating ssDNA binding and ATPase activity.

To test this directly, we performed ssDNA binding assays with Hrq1 and Hrq1ΔN. As previously demonstrated, gel shift assays showed that full-length Hrq1 can bind a poly(dT) ssDNA of 30 nt in length (Fig. 2b). In contrast, however, Hrq1ΔN displayed only modest (<5%) ssDNA binding at the highest concentration of protein tested (5 nM). As with the ATPase assays above, these data again suggest an important role for the Hrq1 N-terminus in ssDNA binding. Indeed, the isolated NTD displayed more ssDNA binding than Hrq1ΔN (Fig. 2c).

By definition, the core motor domains of DNA helicases interact with DNA during unwinding, but the data presented here indicate that Hrq1 has a secondary ssDNA binding site in its NTD that increases the affinity of the protein for its nucleic acid substrate (Fig. 2b and c) and coordinates ssDNA binding with ATP hydrolysis (Fig. 2a). This is similar to the NTD of RECQL4, which is responsible for DNA interactions that drive ssDNA strand annealing and DNA strand exchange (Macris et al. 2006; Xu and Liu 2009; Keller et al. 2014; Rogers and Bochman 2017; Rogers et al. 2019). Although Hrq1 does not have strand annealing or exchange activity (Rogers and Bochman 2017), it may still utilize its NTD—and perhaps the natively disordered portions thereof—to bind to its wide variety of high-affinity substrates like G-quadruplex (G4) structures, D-loops, and RNA–DNA hybrids (Rogers et al. 2017; Nickens et al. 2018). This would again parallel the DNA structure binding displayed by the RECQL4 NTD (Keller et al. 2014; Marino et al. 2016; Papageorgiou et al. 2023).

In vivo roles of the NTD

Fibroblasts from Rothmund-Thomson syndrome patients with *recql4* mutations display sensitivity to drugs that induce DNA inter-strand crosslinks (ICLs) but not other types of DNA damage (Jin et al. 2008), suggesting a role for the helicase in ICL repair. This function appears to be conserved in Hrq1 because we have shown that it functions in the Pso2 ICL repair pathway in yeast (Bochman et al. 2014; Rogers et al. 2014, 2017, 2020a, 2020c). To determine if the Hrq1 NTD is needed for its ICL repair activity, we engineered

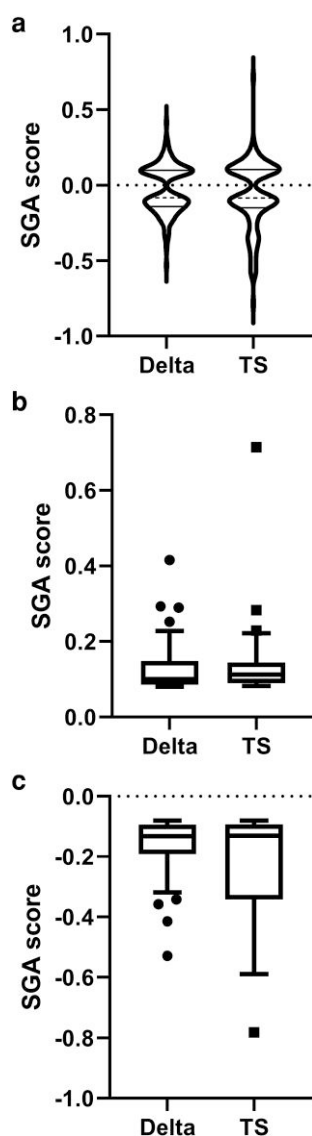


Fig. 4. Analysis of the distribution and magnitudes of the synthetic genetic interactions with *hrq1ΔN*. a) Violin plots of the synthetic genetic interactions with the single-gene deletion collection (Delta) and TS collection. The median values are denoted with dashed lines, and the quartiles are shown as solid lines. The SGA data are also shown in separate box and whisker plots drawn using the Tukey method for the positive (b) and negative (c) interactions with the deletion and TS collections. The individually plotted points outside of the inner fences represent outliers (ie interactions with mutants yielding the strongest SGA scores) and correspond to alleles whose SGA score is greater than the value of the 75th quartile plus 1.5 times the inter-quartile distance (IQR) for positive interactions and alleles whose SGA score is less than the value of the 25th quartile minus 1.5IQR for negative interactions. The significant differences between SGA data sets discussed in the main text were calculated using the Kruskal–Wallis test and Dunn’s multiple comparisons test.

the *hrq1ΔN* allele in *S. cerevisiae* and exposed the cells to the ICL-inducing agent DEB.

As shown in Fig. 3, *hrq1Δ* cells are sensitive to DEB, and the catalytically inactive allele of *HRQ1* (*hrq1*-K318A) displays even greater sensitivity. We have observed this phenomenon before (Bochman et al. 2014; Rogers et al. 2017, 2020a) and note that *Hrq1*-K318A retains DNA binding activity in vitro (Nickens et al. 2018) and is produced at wild-type levels in vivo (Bochman et al. 2014). Thus, we hypothesize that the mutant protein acts as a dominant negative

by being recruited to sites of ICL damage and then blocking compensatory pathways from accessing the lesions for repair. In contrast to *hrq1*-K318A, the *hrq1ΔN* mutant phenocopies the null (Fig. 3). This corresponds with the data presented above, showing that recombinant *Hrq1ΔN* is defective for DNA binding (Fig. 2b and c). We hypothesize that the *Hrq1ΔN* protein does not interact with ICL substrates in vivo, explaining why this allele is as ineffective in ICL repair as the *hrq1Δ* strain.

The genetic interactome of *hrq1ΔN*

To gain further insight into the role(s) of the *Hrq1* NTD in vivo, we performed SGA analysis of the *hrq1ΔN* allele by crossing it to the *S. cerevisiae* single-gene deletion (Giaever and Nislow 2014) and TS mutant collections (Kofoed et al. 2015). Because we previously performed identical analyses for *hrq1Δ* and *hrq1*-K318A strains (Rogers et al. 2020b; Sanders et al. 2020), we can compare the data for differential interactions that may explain the distinct phenotype of our *hrq1* mutants. Here, we found 286 synthetic genetic interactions with *hrq1ΔN* that were nearly evenly split between the deletion (151) and TS collections (135). The split was also similar between negative (81 in the deletion collection, 76 in the TS collection) and positive genetic interactions (70 in the deletion collection, 59 in the TS collection), and both types of interactors are listed in Table 1.

Figure 4 shows the frequency distribution of the SGA scores for the screens of both collections as violin plots (Fig. 4a), with separate box blots for the negative (Fig. 4b) and positive (Fig. 4c) synthetic genetic interactions, where the outliers are denoted with individual points. These outliers are the mutants with the statistically strongest phenotypes. As depicted in Fig. 4 though, the vast majority of the synthetic phenotypes were mild increases or decreases in growth of the double-mutants compared to the parental strains. This is similar to the SGA analyses of *hrq1Δ* and *hrq1*-K318A, as well as their equivalent alleles of the Sgs1 helicase (Sanders et al. 2020). Screening double mutants of *hrq1* and *sgs1* by SGA lead to a synergistic increase in the number of triple mutant strains with strong growth phenotypes, so it would be interesting to also, for instance, cross *hrq1ΔN sgs1Δ* and *hrq1ΔN sgs1*-K706A strains to the deletion collection for analysis.

Hrq1ΔN interactions

Regardless, combining the deletion and TS data, loss of the *Hrq1* NTD yielded strong positive genetic interactions with seven genes: *RPS9B*, *RTG3*, *RTG2*, *YGL214W*, *ARP3*, *LAS17*, and *MPS3*. The *RPS9B* gene encodes a protein found in the small ribosomal subunit (Mager et al. 1997), and *Hrq1* has previously been linked to ribosome assembly and maturation at the genetic and physical interactome levels (Rogers et al. 2020b). *Rtg2* regulates the subcellular localization of *Rtg3* and communicates mitochondrial dysfunction to the nucleus (Liao and Butow 1993; Sekito et al. 2000). *Hrq1* and *RECQ14* are the only known RecQ family helicases with both nuclear and mitochondrial roles, though their functions in mitochondria remain unclear (Croteau et al. 2012; Koh et al. 2015). It is tempting to speculate that *Hrq1* is also involved in this nuclear-mitochondrial communication pathway or acts downstream of the communication to help maintain the stability of the nuclear and/or mitochondrial genomes. *Arp3* and *Las17* are interacting proteins that regulate actin filaments (Machesky and Gould 1999; Madania et al. 1999). Although it is unclear what *Hrq1* may have to do with the cytoskeleton, *arp3* mutation also decreases telomere length (Ungar et al. 2009), and *Hrq1* has multiple links to telomere biology (Bochman et al. 2014; Rogers et al. 2017, Nickens et al. 2018, 2019, 2024; Nickens and Bochman 2021).

Table 2. GO term enrichment of negative genetic interactors with *hrq1ΔN*.

GO term (GO ID)	Genes annotated to the GO term ^a	GO term usage in gene list	Genome frequency of use
transcription by RNA polymerase II (GO:0006366)	<i>CDC73, CSE2, EAF7, IXR1, MCM1, MED7, NAB3, NUT2, POB3, RAD4, ROX3, SGF73, SIN3, SPT3, SRB6, STB5, SWI4, TAF2, TFG2, YBL021C, YGL237C</i>	21 of 138 genes, 15.22%	490 of 6,488 annotated genes, 7.55%
mitotic cell cycle (GO:0000278)	<i>CDC20, ELM1, GIC1, GRR1, LTE1, MCM1, MOB1, MPS1, PEF1, PSE1, SAP190, SIN3, SMC2, SPC29, SWI4, TUB2, YCS4</i>	17 of 138 genes, 12.32%	330 of 6,488 annotated genes, 5.09%
cytoskeleton organization (GO:0007010)	<i>ARP3, BBP1, CDC31, CMD1, COF1, EFB1, ELM1, ENT3, GIC1, LST8, MPS1, NDC1, SPC29, SPC97, TUB2</i>	15 of 138 genes, 10.87%	252 of 6,488 annotated genes, 3.88%
response to chemical (GO:0042221)	<i>ASK10, CMD1, GPR1, GRR1, IXR1, LDB19, STB5, STE50, TIM18, YHK8, YOS9</i>	11 of 138 genes, 7.97%	408 of 6,488 annotated genes, 6.29%
transmembrane transport (GO:0055085)	<i>ASK10, KCH1, MIR1, MUP3, PAM16, PMA2, SEC63, TOM70, YHK8, YME1</i>	10 of 138 genes, 7.25%	278 of 6,488 annotated genes, 4.28%
regulation of cell cycle (GO:0051726)	<i>CDC20, GIC1, GRR1, LTE1, MOB1, MPS1, PSE1, SMC2, SPO16, YCS4</i>	10 of 138 genes, 7.25%	245 of 6,488 annotated genes, 3.78%
organelle fission (GO:0048285)	<i>CDC20, GIC1, LTE1, MOB1, MPS1, PSE1, SMC2, SPO16, TUB2, YCS4</i>	10 of 138 genes, 7.25%	225 of 6,488 annotated genes, 3.47%
meiotic cell cycle (GO:0051321)	<i>CDC20, LGE1, MPS1, POL1, SIN3, SMC2, SPO16, SPO20, TGS1, YCS4</i>	10 of 138 genes, 7.25%	289 of 6,488 annotated genes, 4.45%
cellular respiration (GO:0045333)	<i>KGD2, LPD1, MBR1, OAR1, QCR2, QCR9, YBL021C, YBR238C, YGL237C, YJR120W</i>	10 of 138 genes, 7.25%	86 of 6,488 annotated genes, 1.33%
nucleus organization (GO:0006997)	<i>APQ12, CMD1, CTI6, NDC1, NUP57, SIN3, SMC2, TGS1, YCS4</i>	9 of 138 genes, 6.52%	99 of 6,488 annotated genes, 1.53%

^a Three negative genetic interactors (YML090W, YJL175W, and YMR245W) represent dubious ORFs, and 11 (FPR2, YPR084W, YSC83, DFR1, IRA2, AIM4, YJL213W, KAP114, PAC10, COX10, and PET127) could not be mapped to terms in the current GO set or are annotated to the root node for the set being used. Thus, these 14 genes do not appear in the GO analysis.

Table 3. GO term enrichment of positive genetic interactors with *hrq1ΔN*.

GO term (GO ID)	Genes annotated to the GO term ^a	GO term usage in gene list	Genome frequency of use
mitotic cell cycle (GO:0000278)	<i>ACT1, BRN1, CDC11, CDC48, DIA2, DPB11, MCD1, MPS3, MYO2, PDS5, PFY1, SDA1, SFI1, SMC4, SPT6, STU2, SWI6, VRP1</i>	18 of 113 genes, 15.93%	330 of 6,488 annotated genes, 5.09%
cytoskeleton organization (GO:0007010)	<i>ACT1, ARP3, CDC48, ICE2, LAS17, MPS3, MSB4, MYO2, PFY1, RSP5, SFI1, SHE4, SRV2, STU2, VRP1</i>	15 of 113 genes, 13.27%	252 of 6,488 annotated genes, 3.88%
chromatin organization (GO:0006325)	<i>CSM1, DIA2, GTR1, IES1, MPS3, NHP10, ORC1, ORC2, PDS5, RSP5, RVB2, SMC4, SPT5, SPT6</i>	14 of 113 genes, 12.39%	256 of 6,488 annotated genes, 3.95%
lipid metabolic process (GO:0006629)	<i>ACB1, ARF1, CDC1, DGA1, ERG3, GPI10, GPI12, PBN1, RSP5, SAC1, SUR1, VPS4, YMR134W</i>	13 of 113 genes, 11.50%	296 of 6,488 annotated genes, 4.56%
organelle fission (GO:0048285)	<i>ARF1, BRN1, CSM1, EBP2, MCD1, MPS3, MSH4, PDS5, SMC4, SRV2, SWI6</i>	11 of 113 genes, 9.73%	225 of 6,488 annotated genes, 3.47%
DNA repair (GO:0006281)	<i>ACT1, CDC1, DPB11, MCD1, NHP10, NSE4, PDS5, POL3, RAD27, RPN4, SPT5</i>	11 of 113 genes, 9.73%	269 of 6,488 annotated genes, 4.15%
rRNA processing (GO:0006364)	<i>EBP2, FAL1, RPF2, RPS27B, RPS9B, RSP5, RVB2, SLX9, SPT5, TSR2, TSR3</i>	11 of 113 genes, 9.73%	327 of 6,488 annotated genes, 5.04%
transcription by RNA polymerase II (GO:0006366)	<i>MKS1, RPN4, RSP5, RTG1, RTG2, RTG3, RVB2, SPT5, SPT6, SWI6, YML062C</i>	11 of 113 genes, 9.73%	490 of 6,488 annotated genes, 7.55%
response to chemical (GO:0042221)	<i>CDC48, EOS1, GRX2, GTR1, PBN1, RPN4, RSP5, RTG1, RTG3, YAR1</i>	10 of 113 genes, 8.85%	408 of 6,488 annotated genes, 6.29%
regulation of organelle organization (GO:0033043)	<i>ARF1, CDC48, LAS17, PFY1, RSP5, SEC23, SRV2, SWI6, VRP1</i>	9 of 113 genes, 7.96%	227 of 6,488 annotated genes, 3.50%

^a Eight positive genetic interactors (YML012C-A, YLR338W, YGL024W, YBL094C, YDL062W, YMR031W-A, YMR294W-A, and YGL214W) represent dubious ORFs, and seven (CAT2, YBR016W, ARA2, YHR180W, PDE1, NCE101, and PKP2) could not be mapped to terms in the current GO set or are annotated to the root node for the set being used. Thus, these 15 genes do not appear in the GO analysis.

Therefore, synthetic genetic interaction between *hrq1ΔN* and *arp3Δ* may be related to their roles in telomere maintenance. However, human RECQL4 is a microtubule-interacting protein (Yokoyama et al. 2019), so an interaction between *Hrq1* and the cytoskeleton cannot be ruled out. In support of this is the fact that *Mps3* localizes to telomeres during meiosis, and deletion of

MSP3 likewise synergizes with *hrq1ΔN*. Finally, *YGL214W* is a dubious ORF that likely does not encode a functional protein (<https://www.yeastgenome.org/locus/S000003182>). However, the *ygl214wΔ* allele in the deletion collection also disrupts the nearby *SKI8* gene (Kushner et al. 2003), and *Ski8* is required for DSB repair during meiosis (Arora et al. 2004).

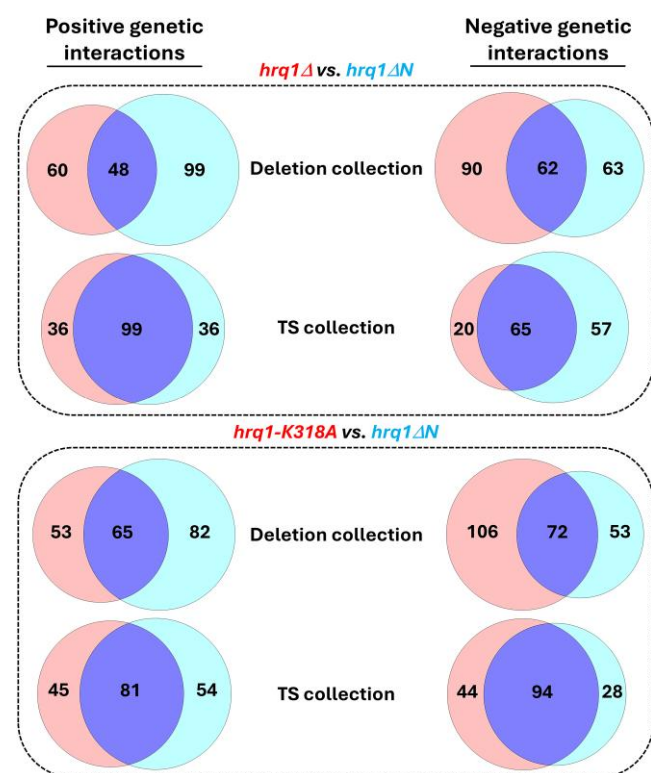


Fig. 5. Venn diagrams of the shared synthetic genetic interactions displayed by *hrq1Δ* and *hrq1ΔN* (top) and *hrq1-K318A* and *hrq1ΔN* (bottom). The numbers of unique and overlapping interactions are labeled. The circles and their percent overlap were drawn to scale using Venn Diagram Plotter software (v. 1.6.7458.27187). Full lists of shared and unique interactions are found in [Supplementary Tables 3 and 4](#).

Similar to the above, the combined SGA data revealed five strong negative genetic interactions (*YSC83*, *SLX9*, *GIC1*, *MRM2*, and *YEF3*), and common themes emerge. For instance, *Slx9* is involved in ribosome biogenesis ([Bax et al. 2006](#)), and *Yef3* is a translation elongation factor ([Chakraborty and Triana-Alonso 1998](#)), so perhaps their double mutants with *hrq1ΔN* are related to *Hrq1*'s putative role in ribosome biology ([Rogers et al. 2020b](#)). That said, *Slx9* is also a G4 structure-binding protein ([Gotz et al. 2019](#)), and G4 DNA is a preferred substrate for *Hrq1* in vitro ([Rogers et al. 2017](#)). Therefore, the negative genetic interaction between *hrq1ΔN* and *slx9Δ* could be due to their roles in DNA secondary structure maintenance. In contrast, *Ysc83* and *Mrm2* are mitochondrial proteins ([Pintard et al. 2002](#); [Sickmann et al. 2003](#)), and the negative genetic interactions here again point toward *Hrq1*'s undefined role(s) in mitochondria. *GIC1* encodes a protein that regulates mitotic exit, with links to regulation of the actin cytoskeleton ([Hofken and Schiebel 2004](#)), but as stated above, how actin is connected to *Hrq1* and its NTD remains unknown.

Cellular pathways affected by *hrq1ΔN*

Next, we used Gene Ontology (GO) term enrichment analysis to determine which pathways in yeast are affected by *hrq1ΔN* and mutation in other genes. [Table 2](#) shows the top 10 GO terms and their associated genes related to negative genetic interactions with *hrq1ΔN*. The GO terms include transcription by RNA polymerase II, regulation of organelle organization, DNA repair, chromatin organization, mitotic cell cycle, peptidyl-amino acid modification, cytoskeleton organization, mitochondrion organization, organelle

fission, and response to chemical. Similarly, [Table 3](#) collates the top 10 GO terms related to positive genetic interactions with *hrq1ΔN* (mitotic cell cycle, cytoskeleton organization, chromatin organization, lipid metabolic process, organelle fission, DNA repair, rRNA processing, transcription by RNA polymerase II, response to chemical, and regulation of organelle organization), which overlap by 80% with the GO terms for negative interactions.

These results make sense with regard to, for instance, *Hrq1*'s connections to transcription ([Rogers et al. 2020b](#)), DNA ICL repair ([Bochman et al. 2014](#); [Rogers et al. 2017, 2020a](#)), and mitochondrial localization ([Koh et al. 2015](#)). Similarly, they reinforce the strong synthetic interactions described above (eg cytoskeleton organization and the synergism between *hrq1ΔN* and *arp3*, *las17*, and *gic1*; [Fig. 4](#)). Therefore, investigating the genetic interactions between *hrq1ΔN* and other *hrq1* alleles should be fertile ground for future research into multiple physiological functions for this RecQ family helicase. However, it should also be noted that a limit of the *Saccharomyces* Genome Database GO Slim Mapper (<https://www.yeastgenome.org/goSlimMapper>) used here is that it does not provide statistical support, such as FDR-adjusted *P*-values, which may limit the interpretation of the results reported above.

How do the synthetic genetic interactions with *hrq1ΔN* compare to those with *hrq1Δ* and *hrq1-K318A*?

Our in vitro biochemistry demonstrates that *Hrq1ΔN* is still a catalytically active protein ([Fig. 2](#)), suggesting that it could perform some of the functions of *Hrq1* in vivo. However, cells expressing *hrq1ΔN* display *hrq1Δ*-levels of DNA ICL sensitivity ([Fig. 3](#)). This begs the question: how do the synthetic genetic interactions with *hrq1ΔN* uncovered here compare to those published for *hrq1Δ* and *hrq1-K318A*? Analyzing all combinations of positive and negative synthetic genetic interactions shows that there is similar overlap between mutants from the deletion and TS collections with all of the *hrq1* alleles investigated ([Fig. 5](#)). In general, the genetic interactions overlapped more with TS alleles, with 53% to 77% of genetic interactions shared between *hrq1Δ* or *hrq1-K318A* and the *hrq1ΔN* data set. The interactions with the deletion collection were more varied, with 33% to 58% overlap comparing *hrq1ΔN* to *hrq1Δ* and *hrq1-K318A*. These data are similar to those for comparisons between *hrq1Δ* and *hrq1-K318A* themselves ([Sanders et al. 2020](#)).

Conclusions and perspectives

Here, we have reported a comprehensive set of synthetic genetic interactions between nearly all of the genes in the *S. cerevisiae* genome and the NTD-truncated *hrq1ΔN* allele of the *Hrq1* helicase, a homolog of disease-linked human RECQL4. The data sets included herein supplement the existing sets of known *hrq1Δ* and *hrq1-K318A* interactions ([Rogers et al. 2020b](#); [Sanders et al. 2020](#)), expanding the genetic interactome landscape of *hrq1*. As with the deletion and catalytically inactive mutants, it is clear that *hrq1ΔN* genetically interacts with genes in multiple important pathways in yeast that underpin the maintenance of genome integrity. As such, these SGA analyses have also helped us to generate testable hypotheses that are driving our ongoing and planned research. As with *hrq1Δ* and *hrq1-K318A*, the *hrq1ΔN* genetic interactome again suggests links to transcription, suggesting that RECQL4 may also functionally interact with the transcription machinery as previously established for human RECQL5 and RNA polymerase II ([Aygün et al. 2008](#); [Izumikawa et al. 2008](#); [Saponaro et al. 2014](#)). Indeed, we are currently investigating potential roles for *Hrq1* in transcription

and are establishing tools to extend this work to the *Drosophila melanogaster* RECQL4 homolog (Capp et al. 2009). Given that the role of Hrq1 in ICL repair is extensively studied, ongoing work is also screening the double mutants generated here and previously (Rogers et al. 2020b; Sanders et al. 2020) to identify additional pathways involved in ICL repair. It is our hope that the data provided here will spur additional work in the field on RecQ4 subfamily helicases to finally elucidate the roles of these enzymes in genome integrity.

Data availability

The data underlying this article are available in the article or in its online [Supplementary material](#). [Supplementary Figs. 1–3](#) show AlphaFold structures of RecQ4 family helicases, Hrq1 disorder prediction, and confirmation of SGA hits, respectively. [Supplementary Tables 1–4](#) include the AlphaFold inputs sequences, *hrq1ΔN* SGA data, a comparison of *hrq1Δ* and *hrq1ΔN* genetic interactions, and a comparison of *hrq1-K318A* and *hrq1ΔN* genetic interactions.

Supplemental material available at [G3](#) online.

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Conflicts of interest

None declared.

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